

Workshop 1.5: Hidden Assumptions (with answers)

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Workshop Structure

As always, we are using the workflow below to structure the workshops:

1. Formulate a research question.
2. Perform exploratory data analysis.
3. Identify any hidden assumptions [Focus of today].
4. Fit an appropriate model.
5. Diagnose the model and check assumptions.
6. Summarise the results.
7. Interpret and provide inferences.

Learning Objectives

1. Learn to identify potential issues (or “hidden assumptions”) without seeing the data.
2. Identify potential hidden assumptions in a real world dataset.
3. Use R to perform simple descriptive summaries of the data to assist in this.

These skills will be used throughout the course.

Task 1

In this workshop, you will come up with some examples of various datasets that are likely to violate a hidden assumption. Some of these are related to residual error, which we will cover in more detail next week. For now, understand residual error as the difference between what a model predicts and the actual observation. For example, if a model predicts a flat costs £850 to rent each month, but the actual rent was £800, then the residual error is observed minus predicted (or $Residual = y - \hat{y}$). For the flat example, the residual error would be $800 - 850 = -50$, or £-50 difference.

These assumptions (with some simple examples) are:

Assumption	Description	Example of Violation
1. Validity	The data is able to answer the research question effectively.	Using IQ to measure overall intelligence (what does IQ even measure?).
2. Representativeness	The data reflects the population or group being studied.	Using survey data from a rural area to draw conclusions about an urban population.
3. Additivity	Effects of predictors are additive and there are no interactions between them.	Assuming that study hours and sleep hours independently affect exam scores without considering their combined effect (imagine a Venn diagram with study and sleep overlapping).
4. Linearity	The relationship between predictors and the outcome is linear.	Assuming plant growth will continue to increase indefinitely without ever levelling out.
5. Independence of Errors	Observations are independent of each other. (Related to residual errors.)	If modelling height, the data come from the same group of children over time.
6. Equal Variance of Errors	Residuals have constant variance across all levels of predictors ("homoscedasticity"). (Related to residual errors.)	If modelling income, variability in salaries is not the same across all levels of a company's hierarchy.
7. Normality of Errors	Residuals should be normally distributed. (Related to residual errors.)	If modelling income, the presence of any extremely wealthy individuals will be hard for the model to understand and explain.

For this exercise, form groups of two to four at your table. Your task is to come up with seven datasets that you believe are likely to violate assumptions one to seven in the above table. Once you have come up with an example for a given assumption, choose one of your group to upload your example the MyAberdeen discussion board, such that we build up a "portfolio" of lousy datasets.

If it helps, you can think of this exercise as essentially asking you to be a terrible scientist. For each assumption, think of an experiment or how you might collect data in such a way as to

violate one of these assumptions. You might decide to focus on trying to come up with a lousy experiment to understand, which you do in the worst possible way:

- Ostrich wing length (e.g. “we measured wing length by looking at the wing through binocs and guessing to the nearest mm” - **Validity**)
- European otter density (e.g. “we measured the density of European otters in Tokyo zoo to understand otter dynamics in Spain” - **Representativeness**)
- Plant disease occurrence (e.g. “we assume residual error [difference between prediction and truth] of number of pathogens will be constant as a plant becomes older” - **Equal Variance of Errors**)
- Dolphin weight (e.g. “we assume that dolphins will continue to grow at the same rate throughout their lives” - **Linearity**)
- Number of corrupt politicians (e.g. “we assume most politicians are similar and that it’s not just one or two that are extremely corrupt” - **Normality of Errors**)

You’re by no means limited to the silly examples above. Feel free to think of your own silly (or not) example datasets/context.

To ensure that we have examples for each assumption, within your group, have one person run the following code to determine the order of assumptions (from 1. **Validity** to 7. **Normality of Errors**) that you will work through:

```
# The code below is essentially a lottery machine
# x = are the options you want to be selected from (here "1 to 7" or 1:7)
# size = the number of draws you want made
# replace = FALSE determines if, once one of x has been selected,
# if it can still be selected in subsequent draws
sample(x = 1:7, size = 7, replace = FALSE)
```

If you are unsure what to do, please ask a member of staff, and also discuss your ideas with them during this first task.

Task 2

Still within your groups, visit the following [NBN atlas](#) web page. This page contains all recordings of red deer in the UK. Spend some time exploring this page, as well as finding out how the data is collected.

Question Set One

1. Using this dataset, consider which assumptions it is most likely to violate.

```
# This dataset almost certainly violates the assumptions of validity and representativeness.
```

```
# Validity:
```

```
# For each data point, an animal had to be present in that location and also *seen* by the observer. Given we cannot guarantee all observers see an animal
```

, given it was present, and it's unlikely everyone has the same level of skill, it seems likely that this will vary dramatically.

Additionally, are we sure that the person saw a red deer? Maybe it was a roe deer, a dog, or even Bigfoot.

Representativeness:

The data only contains instances where someone has seen and reported a red deer. It doesn't include observations when someone tried to spot a deer but failed. This means that all of the blank areas on the map have two meanings: someone tried to spot a deer but failed, or no one tried to spot a deer. So we have highly ambiguous data. I.e. in any area with no deer sightings, are you sure there aren't any deer there?

Additionally, are we sure all areas of the UK are surveyed with the same intensity? Are people equally likely to report a red deer sighting if they're close to Fort William, as they are in London? I can easily imagine someone living near Fort William being fairly accustomed to red deer, whereas someone spotting a deer in London would be much more exciting and therefore be more likely to be reported.

In short, these kinds of datasets have limited value because what the data actually captures is: Observations where a deer was present, seen, and reported. But what we want it to represent is: Observations where deer are either present or absent.

Having done this, compare the [NBN atlas](#) with the [IUCN](#).

Question Set Two

1. How reliable do you think the IUCN geographic range is for the UK?

The core of the range is likely reliable, but the range edge is likely highly unreliable, with some areas probably overestimated and others underestimated. The reason for this, is that as we approach a species range edge, we might find 1) the species are less common so are not seen as often, 2) they are relatively "new" at the range edge so are not reported as often, 3) they might colonise an area but quickly go extinct [meaning we might have false positive "detections"], 4) plus lots of additional "little" things that all contribute to the species "detections" being highly unreliable, especially at range edges.

2. Do you think the non-UK geographic range is as reliable or less so?

The UK public tends to be quite involved with reporting sightings, particularly with birds. In other countries, this culture either doesn't exist or is less pronounced, or it may be more pronounced than in the UK (e.g. in some datasets, it's not uncommon for all observations from a single country to be from a single highly motivated individual). As a result, it seems very likely that the accuracy of the range, across multiple countries, is going to be highly variable (and not particularly informative) as we move from "culture" to "culture".

3. Do you think there is any value in the IUCN geographic range for decision making?

There is, but in a limited capacity. It's useful for very broad decision making - decisions that rely on e.g. "Red deer are present in country X" - but is very likely to be detrimental for any fine-scale decision making - decisions that rely on "Red deer are expanding into neighbourhood Y of city X."

That's often the case in science - results may still be useful for very broad level understanding if the underlying data is of "low quality", but if we want to make more precise and accurate decisions, we really need to data to be "top quality".

Task 3

Return to the `alligator_bite.txt` dataset from the first workshop. We did not give you any information as to *how* the data were collected, so gauging how likely it was to meet or violate assumptions is difficult (but not impossible). To allow you to do so now, here is a description of data collection:

Each alligator was retrieved from a communal pond containing 100 alligators. Although the alligators were not individually tagged, farm staff were able to identify each one, ensuring that observations were independent. After retrieval, the alligator was blindfolded and moved to a safe and secure location. Once secured, the alligator was restrained by up to three technicians while its length was measured. To measure the length, two additional technicians used a tape measure from the tip of the nose to the tip of the tail. The tape was passed under the restraining technicians when possible to ensure accurate measurements. Lengths were recorded in centimetres, with one decimal point.

After measuring length, the blindfold was removed, and a modified gnathodynamometer was used to measure bite force in pounds per square inch. Since alligators may not immediately bite the gnathodynamometer, the device was placed in the alligator's mouth for up to 60 seconds to capture an appropriate bite strength. Once all measurements were taken, the alligator was returned to the pond before the next individual was retrieved.

Question Set Three

1. With the alligator dataset and data collection description, state which assumptions you believe may be most likely to be violated.

The issues that jump out are:

1. It seems likely that they could sample the same alligators, given they return them to the pond before retrieving the next, even with assurances that they believe this is not the case. If this has occurred, then they have extreme Lack of independence, as they will have repeated measures of the same alligator for some observations.

2. It sounds like actually getting alligators to bite the device might have been a bit hard. Raises the question if we can be sure that each alligator bi

t the device as hard as possible, or just hard enough for the staff to feel like they had a measurement. We could formalise this as the assumption of Validity.

3. The fact that they had to pass the measuring tape in and around technicians to measure the alligators makes it sound like these won't be accurate. This isn't even thinking about what happens if the alligator is moving around while they were trying to measure it. Seems like length wouldn't be measured especially accurately, never mind to one decimal point (which seems absurdly accurate given the difficulties in measuring).

2. Describe how the data collection could have been improved on such that the assumptions would be more likely to be met.

The easiest option may be to have some method to identify which alligators had already been measured. Something like a non-toxic paint mark on their head. That would remove the chance that a previously measured alligator was repeatedly measured.

In terms of getting more accurate measurements, it's a bit harder. Measuring an alligator's length and bite strength will probably always be difficult and the way they do it seems pretty sensible from the little I know about measuring alligator bite strength (which isn't much). Personally, I can't think of a way to resolve this issue, so I would likely look to resolve the measurement error in the model (note, we don't teach you how to do this).

The broader point with this example is that sometimes you can't improve a dataset. In such cases, the important thing is to identify what impacts this will have on the results, which we'll uncover as the course progresses.